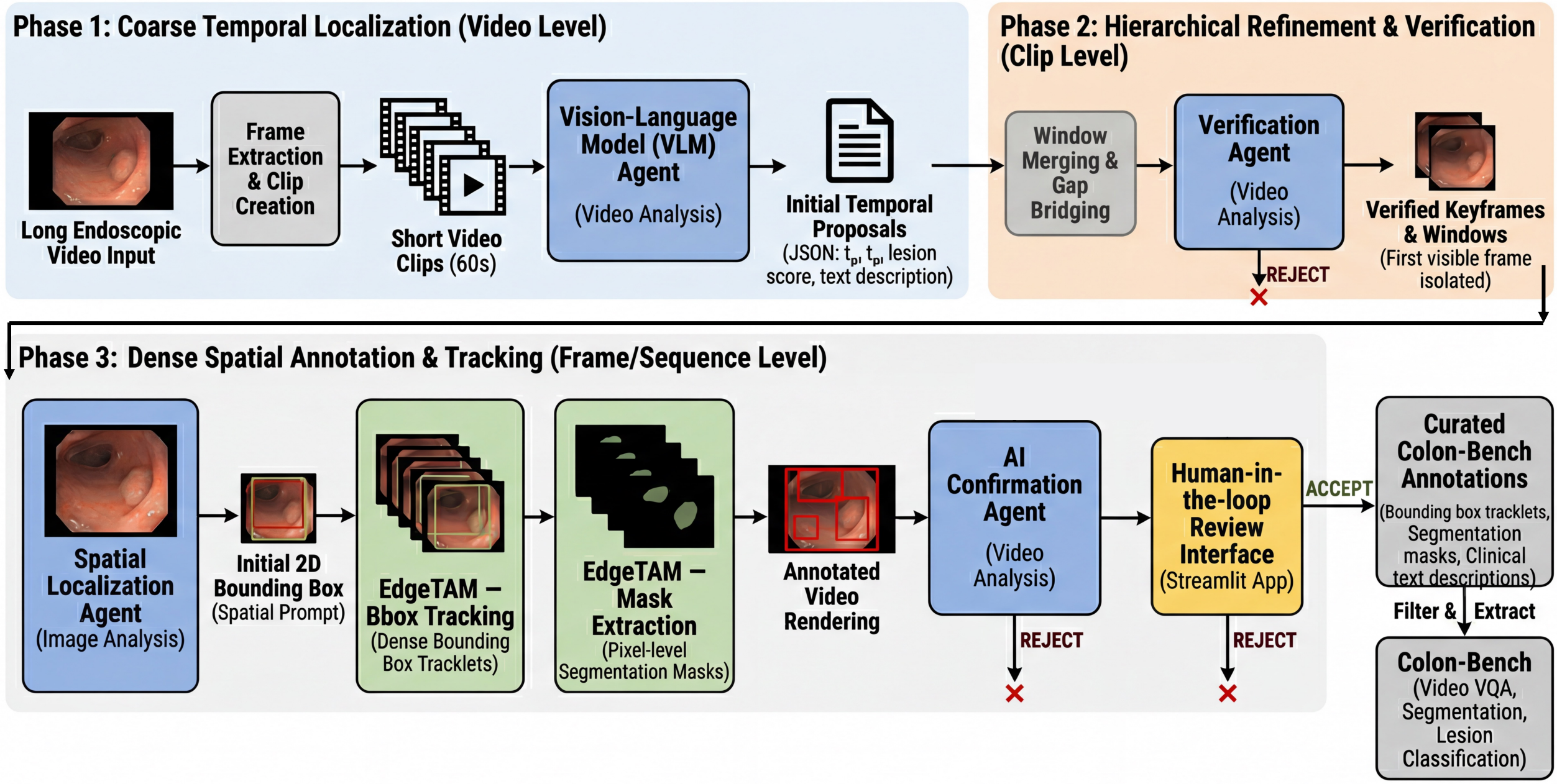
Phase 1: Coarse Temporal Localization (Video Level)
Long Endoscopic Video Input
Frame Extraction & Clip Creation
Short Video Clips (60s)
Vision-Language Model (VLM) Agent
(Video Analysis)
Initial Temporal Proposals
(JSON: t_p, t_p, lesion score, text description)
Phase 2: Hierarchical Refinement & Verification (Clip Level)
Window Merging & Gap Bridging
Verification Agent
(Video Analysis)
REJECT
Verified Keyframes & Windows
(First visible frame isolated)
Phase 3: Dense Spatial Annotation & Tracking (Frame/Sequence Level)
Spatial Localization Agent
(Image Analysis)
Initial 2D Bounding Box
(Spatial Prompt)
EdgeTAM – Bbox Tracking
(Dense Bounding Box Tracklets)
EdgeTAM – Mask Extraction
(Pixel-level Segmentation Masks)
Annotated Video Rendering
AI Confirmation Agent
(Video Analysis)
REJECT
Human-in-the-loop Review Interface
(Streamlit App)
REJECT
ACCEPT
Curated Colon-Bench Annotations
(Bounding box tracklets, Segmentation masks, Clinical text descriptions)
Filter & Extract
Colon-Bench
(Video VQA, Segmentation, Lesion Classification)